

# Online Appendix to “Optimal Disclosure Windows”

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## OA.1 Benchmark: Perfectly Informed Agent

As a benchmark, consider the scenario in which the agent is informed of whether the state is  $\theta = 0$  or  $\theta = 1$  at time 0. As disclosure starts later, the stopping game converges to the benchmark case where the agent is perfectly informed. As mentioned in [Proposition 3](#), the waiting time in the continuation stopping game is bounded above by  $\bar{w}^*$ , which is the waiting time in this benchmark. [Proposition OA.1](#) formalizes this argument (the characterization of  $\bar{w}^*$  is relegated to the Online Appendix).

**Proposition OA.1.** *In the continuation game, there exists  $0 < \bar{w}^* < \infty$  such that the informed agent randomizes over stopping times in  $[0, \bar{w}^*]$  and uninformed agent stops with probability 1 at  $\bar{w}^*$ .*

The proof is the same as the proof of [Theorem 1.B](#) (ii) as is thus omitted.

## OA.2 Omitted Proofs for [Section 3](#)

### OA.2.1 Proof of Lemma 19

Take the derivative of  $V^*(t)$  with respect to  $t$ ,

$$V^{*'}(t) = \frac{e^{\lambda t - r w^*(t)} \lambda \mu (1 - \mu) - e^{\lambda t - r w^*(t)} \mu (r(1 - \mu) + e^{\lambda(t + w^*(t))} (r + \lambda) \mu) w^{*'}(t)}{(1 + (e^{\lambda(t + w^*(t))} - 1) \mu)^2}.$$

It suffices to show

$$e^{\lambda t - r w^*(t)} \lambda \mu (1 - \mu) - e^{\lambda t - r w^*(t)} \mu (r(1 - \mu) + e^{\lambda(t + w^*(t))} (r + \lambda) \mu) w^{*'}(t) < 0,$$

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which is equivalent to showing

$$w^{*'}(t) > \frac{\lambda(1-\mu)}{r(1-\mu) + e^{\lambda(t+w^*(t))}(r+\lambda)\mu}.$$

Recall that

$$w^{*'}(t) = \frac{\beta e^{(\beta+r)w^*(t)} + r}{\beta(e^{(\beta+r)w^*(t)} - 1)} \frac{\lambda(1-\mu)}{r(1-\mu) + e^{\lambda(t+w^*(t))}(r+\lambda)\mu}.$$

The first term is greater than 1 and the second term is positive. The result follows.

### OA.2.2 Proof of Lemma 20

I prove the case for the uninformed agent, the informed agent follows from the same algebra.

By definition,  $Y(t|t_0)$  is given by

$$\begin{aligned} Y(t|t_0) = & (1 - \rho(t_0)) \int_{t_0}^t \lambda e^{-\lambda(s-t_0)} e^{-r(s-t_0)} V^*(s) ds \\ & + ((1 - \rho(t_0))e^{-\lambda(t-t_0)} + \rho(t_0)) e^{-r(t-t_0)} U^*(t). \end{aligned}$$

Take the derivative with respect to  $t$ ,

$$\begin{aligned} Y'(t|t_0) = & e^{-r(t-t_0)} \frac{e^{r(t-t_0)} \rho(t)}{(1 - \rho(t_0))e^{-\lambda(t-t_0)} + \rho(t_0)} \\ & \cdot (\rho'(t) (V^*(t) - U^*(t)) + \rho(t) (U^{*'}(t) - rU^*(t))). \end{aligned}$$

where the first line is strictly positive and the second line is proportional to  $y(t)$ . The result follows.

### OA.2.3 Proof of Lemma 21

Take the derivative of  $Y(t)$  with respect to  $t$ . After simplifying, the derivative is proportional to

$$\lambda(1-\mu)e^{-\lambda t} (V^*(t) - U^*(t)) + ((1-\mu)e^{-\lambda t} + \mu) (U^{*'}(t) - rU^*(t)).$$

Recall that the derivative of the best-case scenario payoff  $Y_0(t)$  is  $\rho'(t) - r\rho(t)$ , which, by assumption, is negative. Because  $V^*(t) - U^*(t) < 0$ , that is, the informed agent's equilibrium continuation payoff is lower than the uninformed, it suffices to show if  $\rho'(t) - r\rho(t) < 0$  for all  $t$ ,  $U^{*'}(t) - rU^*(t) < 0$  for all  $t$ .

The rest of the proof proceeds by brute force calculations. One can explicitly write down the expression for  $U^{*'}(t) - rU^*(t)$ . After simplifying,

$$\begin{aligned}
U^{*'}(t) - rU^*(t) = & \frac{e^{-(\beta+r)w^*(t)}}{(\beta - \lambda)(\rho(t)(e^{\lambda w^*(t)} - 1) + 1)} r\rho(t) (\lambda\rho(t)e^{\lambda w^*(t)} - e^{\beta w^*(t)}(\beta - \lambda + \lambda\rho(t))) \\
& + \rho'(t) \frac{e^{-(\beta+r)w^*(t)}}{(\beta - \lambda)(\rho(t)(e^{\lambda w^*(t)} - 1) + 1)^2} \\
& \cdot (\lambda\rho(t)(e^{\beta w^*(t)} - e^{\lambda w^*(t)})(1 + \rho(t)e^{\lambda w^*(t)} + 1 - \rho(t)) + (\beta - \lambda)e^{\beta w^*(t)}) \\
& + w^{*'}(t) \frac{e^{-rw^*(t)}\rho(t)}{(\beta - \lambda)(\rho(t)(e^{\lambda w^*(t)} - 1) + 1)^2} \\
& \cdot \left( r(\lambda - \beta) + \rho(t)(r(\beta - 2\lambda + \lambda\rho(t))) + \rho(t)\lambda e^{(\lambda-\beta)w^*(t)}(\beta - \lambda + r) \right. \\
& + \lambda e^{(\lambda-\beta)w^*(t)}\rho(t)^2(\lambda - \beta - r - (\lambda + r)e^{\beta w^*(t)} + re^{\lambda w^*(t)} + \beta e^{\lambda w^*(t)}) \\
& \left. + e^{(\lambda-\beta)w^*(t)}\rho(t)(\lambda - \beta)(\lambda + r)e^{\beta w^*(t)} \right)
\end{aligned}$$

The expression for  $w^{*'}(t)$  is given by (23). It can be verified that for each  $t$ , whenever  $\rho'(t) - r\rho(t) < 0$ , the right-hand side is negative for all  $w^*$ . So with the equilibrium waiting time  $w^*(t)$ ,  $U^{*'}(t) - rU^*(t)$ .

## OA.3 Proofs for Section 5

### OA.3.1 Proof of Proposition 4

Let  $q_D(w)$  denote the decision maker's belief that  $\theta = 1$  conditional on no signals arriving in  $[0, w]$ . By Bayes' rule,

$$q_D(w) = \frac{\eta}{\eta + e^{-\lambda w}(1 - \eta)}. \quad (\text{OA.1})$$

Given this belief  $q_D(w)$ , from the perspective of the beginning of the continuation stopping game, denote the decision maker's expected payoff from stopping at  $w$  by  $D(w)$ , then

$$\begin{aligned}
D(w) = & (1 - \eta) \int_0^w e^{-rs} \lambda e^{-\lambda s} e^{-\beta s} ds \\
& + (1 - \eta) \int_0^w e^{-rs} e^{-\lambda s} \beta e^{-\beta s} (1 - (q_D(s) - 0)^2) ds \\
& + (1 - \eta) e^{-rw} e^{-\lambda w} e^{-\beta w} (1 - (q_D(w) - 0)^2) \\
& + \eta \int_0^w e^{-rs} \beta e^{-\beta s} (1 - (q_D(s) - 1)^2) ds \\
& + \eta e^{-rw} e^{-\beta w} ((1 - (q_D(w) - 1)^2)).
\end{aligned} \quad (\text{OA.2})$$

Take the derivative of  $D(w)$  with respect to  $w$ . After simplifying,

$$D'(w) \propto \lambda(1 - q_D(w))q_D(w)^2 - r(1 - (1 - q_D(w))q_D(w)). \quad (\text{OA.3})$$

Define

$$R(w; \eta) := \left( \frac{e^{-\lambda w}(1 - \eta) + \eta}{e^{-\lambda w}(1 - \eta)} + \left( \frac{\eta + e^{-\lambda w}(1 - \eta)}{\eta} \right)^2 \right)^{-1}. \quad (\text{OA.4})$$

Rearrange the right-hand side of (OA.3) and substitute (OA.1) for  $q_D(w)$ ,  $D'(w) \leq 0$  if and only if  $R(w; \eta) \leq r/\lambda$ .

*Claim* OA.1. For any  $w$  and  $\eta$ ,  $\partial R(w; \eta)/\partial w$  and  $\partial R(w; \eta)/\partial \eta$  have the same sign.

*Proof.* To facilitate the proof, note that  $q_D(w)$  depends on both  $w$  and  $\eta$ . With a slight abuse of notation, denote (OA.1) by  $q_D(w; \eta)$ . Then  $R(w; \eta)$ , defined in (OA.5), can be written as

$$R(w; \eta) = \left( \frac{1}{1 - q_D(w; \eta)} + \frac{1}{q_D(w; \eta)^2} \right)^{-1}. \quad (\text{OA.5})$$

Then

$$\frac{\partial R(w; \eta)}{\partial w} = \frac{\partial R(w; \eta)}{\partial q_D} \frac{\partial q_D(w; \eta)}{\partial w} \quad \text{and} \quad \frac{\partial R(w; \eta)}{\partial \eta} = \frac{\partial R(w; \eta)}{\partial q_D} \frac{\partial q_D(w; \eta)}{\partial \eta}.$$

By (OA.1),  $\partial q_D(w; \eta)/\partial w > 0$  and  $\partial q_D(w; \eta)/\partial \eta > 0$ . The result follows.  $\square$

*Claim* OA.2.  $R(w; \eta)$  has the following properties.

(i) For all  $0 < \eta < 1$ ,  $R(w; \eta)$  is single-peaked in  $w$ : there exists a unique  $w_R^*(\eta) = \arg \max_w R(w; \eta)$  with  $w_R^*(\eta) \geq 0$  such that  $R(w; \eta)$  is strictly increasing in  $w$  for  $w \leq w_R^*(\eta)$  and strictly decreasing in  $w$  for  $w > w_R^*(\eta)$ .

(ii) There exists a unique  $0 < \eta_R < 1$  such that

- (a) for all  $\eta < \eta_R$ ,  $w_R^*(\eta) > 0$ ,  $w_R^{*'}(\eta) < 0$ , and  $R(w_R^*(\eta); \eta)$  is constant in  $\eta$ ;
- (b) for all  $\eta \geq \eta_R$ ,  $w_R^*(\eta) = 0$  and  $R(w_R^*(\eta); \eta) = R(0; \eta)$  is decreasing in  $\eta$ ;
- (c) Moreover,  $w_R^*(\eta)$  is continuous for all  $0 < \eta < 1$ . That is,  $\lim_{\eta \rightarrow \eta_R} w_R^*(\eta) \rightarrow 0$ .

(iii) For all  $0 < \eta < 1$ ,  $\lim_{w \rightarrow \infty} R(w; \eta) \rightarrow 0$ .

*Proof.* These properties follow directly from operating on (OA.4).

Specifically, for (ii) (a) and (c), if  $w_R^*(\eta)$  is interior, namely,  $w_R^*(\eta) > 0$ ,  $w_R^*(\eta)$  has a closed form which can be obtained by solving  $\partial R(w; \eta)/\partial w = 0$ . It follows from the functional form of  $w_R^*(\eta)$  that  $w_R^{*'}(\eta) < 0$  and  $\lim_{\eta \rightarrow \eta_R} w_R^*(\eta) \rightarrow 0$ . By definition,

$$\frac{dR(w_R^*(\eta); \eta)}{d\eta} = \frac{\partial R(w_R^*(\eta); \eta)}{\partial \eta} + \frac{\partial R(w_R^*(\eta); \eta)}{\partial w} \frac{dw_R^*(\eta)}{d\eta}.$$

The second term on the right-hand side is zero because  $\partial R(w_R^*(\eta); \eta)/\partial w = 0$ . By [Claim OA.1](#), the first term  $\partial R(w_R^*(\eta); \eta)/\partial \eta = 0$ . Therefore,  $dR(w_R^*(\eta); \eta)/d\eta = 0$ , which means  $R(w_R^*(\eta); \eta)$  is constant in  $\eta$ .

For (ii) (c), it follows from the definition ([OA.4](#)) that  $R(0; \eta) = \left(\frac{1}{1-\eta} + \frac{1}{\eta^2}\right)^{-1}$  and is decreasing in  $\eta$ .  $\square$

**Part (i).** It follows from [Claim OA.2](#) (ii) that  $R(0; \eta_R) = \max_{\eta \in (0,1)} R(w_R^*(\eta); \eta)$ . Denote this maximum value by

$$\Delta := R(0; \eta_R).$$

Then for all  $w \geq 0$  and  $0 < \eta < 1$ ,  $R(w; \eta) \leq R(0; \eta_R) = \Delta$ . This means if  $\Delta \leq r/\lambda$ ,  $R(w; \eta) \leq \Delta$ , which means  $D'(w) < 0$ . (In fact, for any parameters,  $\Delta$  is a constant number is approximately 0.1916.)

**Part (ii).** Fix  $r/\lambda < \Delta$ . The shape of  $D(w)$  depends on  $\eta$ . There are three cases.

**Case 1.** By [Claim OA.2](#), for all  $\eta \geq \eta_R$ ,  $R(w; \eta)$  is decreasing in  $w$  for all  $w \geq 0$  and is thus maximized at  $w = 0$ . Because  $R(0; \eta)$  is decreasing in  $\eta$ , there exists  $\bar{\eta} > \eta_R$  such that  $R(0; \bar{\eta}) = r/\lambda$  and  $R(0; \eta) < r/\lambda$  (and therefore  $R(w; \eta) < r/\lambda$ ) for all  $\eta > \bar{\eta}$ . This implies  $D'(w) \leq 0$  for all  $w \geq 0$  and is thus maximized at  $w = 0$  for all  $\eta \geq \bar{\eta}$ .

**Case 2.** By [Claim OA.2](#), for all  $\eta < \eta_R$ ,  $R(w; \eta)$  is single-peaked in  $w$  and is maximized at  $w_R^*(\eta) > 0$ . Because  $R(0; \eta)$  is decreasing in  $\eta$ , there exists  $\tilde{\eta} < \eta_R$  such that there exists a unique  $w_{DM}^*(\eta)$  where  $R(w_{DM}^*(\eta); \eta) = r/\lambda$ ,  $R(w; \eta) < r/\lambda$  for  $w < w_{DM}^*(\eta)$ , and  $R(w; \eta) > r/\lambda$  for  $w > w_{DM}^*(\eta)$ . This implies  $D(w)$  single-peaked in  $w$  is uniquely maximized at  $w = w_{DM}^*(\eta) > 0$  for  $\eta \in [\tilde{\eta}, \bar{\eta})$ .

**Case 3.** For all  $\eta < \tilde{\eta}$ ,  $R(0; \eta) < r/\lambda$  and  $R(w; \eta)$  intersects with the line  $r/\lambda$  twice. Denote these two intersections by  $w_L(\eta)$  and  $w_H(\eta)$  with  $w_L(\eta) < w_H(\eta)$ . That is,  $R(w_L(\eta); \eta) = R(w_H(\eta); \eta) = r/\lambda$ .

Moreover,  $R(w; \eta) < r/\lambda$  for  $w < w_L(\eta)$  and  $w \geq w_H(\eta)$ , and  $R(w; \eta) > r/\lambda$  for  $w \in [w_L(\eta), w_H(\eta))$ . This means that  $D(w)$  decreases for  $w \in [0, w_L(\eta))$ , increases for  $w \in [w_L(\eta), w_H(\eta))$ , and decreases for  $w \geq w_H(\eta)$ . Therefore, for each  $\eta < \tilde{\eta}$ ,  $D(w)$  has two local maxima, one at  $w = 0$  and one at  $w = w_H(\eta)$ . To determine which is the global maximizer, one can obtain  $D(w_H(\eta))$  by plugging in the closed form expression for  $w_H(\eta)$ , and compare it with  $D(0) = 1 - (1 - \eta)\eta$ . It can be shown that there exists  $\underline{\eta} < \tilde{\eta}$  such that  $D(w_H(\eta)) < 1 - (1 - \eta)\eta$  for all  $\eta < \underline{\eta}$ , and  $D(w_H(\eta)) \geq 1 - (1 - \eta)\eta$  for all  $\eta \in [\underline{\eta}, \tilde{\eta})$ . Then the optimal waiting time  $w_{DM}^*(\eta) = 0$  for  $\eta < \underline{\eta}$  and  $w_{DM}^*(\eta) = w_H(\eta)$  for  $\eta \geq \underline{\eta}$ .

This concludes the proof. As an illustration, **Figure OA.1** plots the function  $D(w)$  for several values of  $\eta$ .

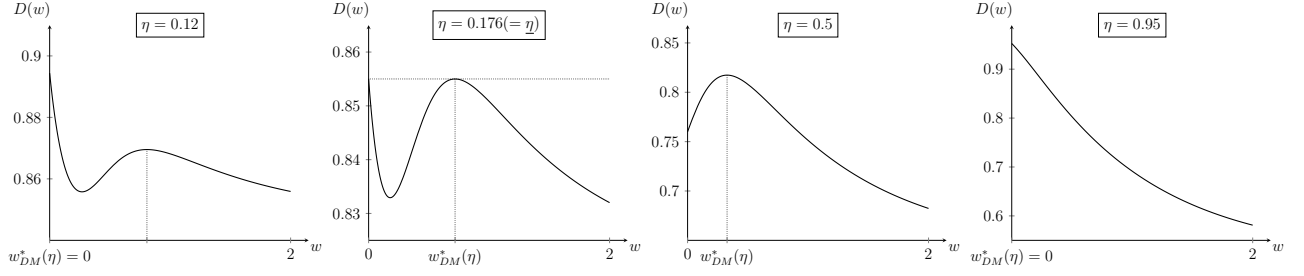


Figure OA.1: Decision maker's expected payoff from stopping at  $w$   $D(w)$  for  $r = 0.5$ ,  $\lambda = 5$ , and  $\beta = 0.5$ . From left to right, the values of  $\eta$  are respectively  $\eta = 0.12, 0.176, 0.5, 0.95$ .